



BEAD FUNDING CYBER PLAN REQUIREMENTS / CISA SECURE BY DESIGN SOFTWARE ACQUISITION GUIDE (SAG)

Unrestricted distribution and usage rights granted by Business Cyber Guardian TM

TOPICS

- What is BEAD
- State Funding Plans
- What are the BEAD Cyber Plan Requirements
- NTCA Guidance
- What is the CISA Software Acquisition Guide (SAG)
- Who created the SAG materials
- How do Product Manufacturers use the SAG
- How do Consumers use the SAG to assess a product for Secure by Design Best Practices
- How can BEAD Implementers use NTCA and CISA Guidance
- Resources

NOTICE OF FUNDING OPPORTUNITY

BROADBAND EQUITY, ACCESS, AND DEPLOYMENT PROGRAM

EXECUTIVE SUMMARY

A. Federal Agency Name

National Telecommunications and Information Administration (NTIA), U.S. Department of Commerce

B. Funding Opportunity Title

Broadband Equity, Access, and Deployment Program

C. Announcement Type

Initial

D. Funding Opportunity Number

NTIA-BEAD-2022

E. Assistance Listing (CFDA Number)

11.035

[BEAD Program Website](https://broadbandusa.ntia.gov/funding-programs/policies-waivers/BEAD_Restructuring_Policy_Notice)

https://broadbandusa.ntia.gov/funding-programs/policies-waivers/BEAD_Restructuring_Policy_Notice

WHAT IS THE BEAD PROGRAM



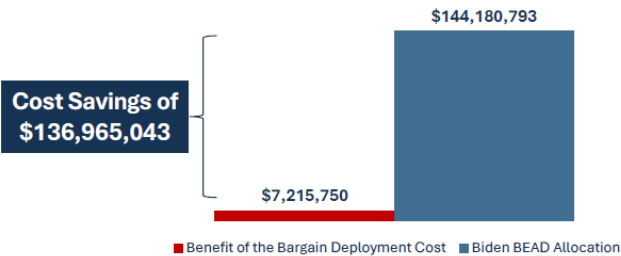
- \$42.45 billion federal grant program that aims to connect every American to high-speed internet
- States and territories can use their allocated funds for these eligible uses:
 - Planning for the deployment of high-speed internet, including conducting research, collecting data, outreach, and training
 - Deploying or upgrading internet infrastructure in unserved or underserved areas, or improving service to community anchor institutions
 - Installing internet and Wi-Fi service in multi-unit residential buildings
 - Internet access, adoption and use programs
 - Workforce readiness programs to meet the needs of the infrastructure programs
 - Other activities that increase meaningful use of constructed infrastructure, at the discretion of the Assistant Secretary

STATE FUNDING PLANS

BEAD Final Proposal Connecticut Overview

Total Allocation: \$144,180,793
Total Deployment Cost: \$7,215,750
of Eligible Broadband Serviceable Locations: 1,782

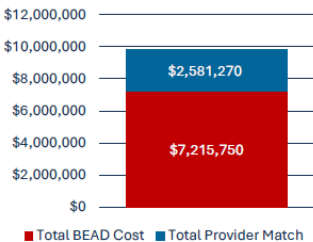
Benefit of the Bargain Saves Taxpayer Money:



Technology Mix Reflects the Competitive Marketplace:

Technology Type	# of Locations	% of Locations
Fiber-to-the-Premises	1,396	78.3%
Hybrid Fiber-Coaxial	386	21.7%
All Technologies	1,782	100%

BEAD Outlay Reduced Due to Provider Cost Sharing:



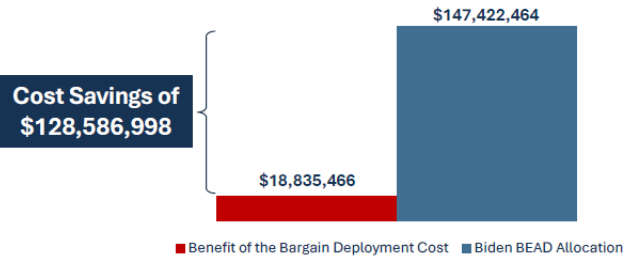
Avg. BEAD Cost per Location: \$4,049
Avg. Match per Location: \$1,449

Note that monetary values shown on this page are rounded to the nearest dollar.

BEAD Final Proposal Massachusetts Overview

Total Allocation: \$147,422,464
Total Deployment Cost: \$18,835,466
of Eligible Broadband Serviceable Locations: 3,808

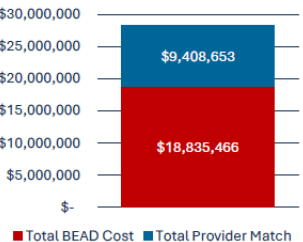
Benefit of the Bargain Saves Taxpayer Money:



Technology Mix Reflects the Competitive Marketplace:

Technology Type	# of Locations	% of Locations
Hybrid Fiber-Coaxial	1,852	48.6%
Low Earth Orbit Satellite	1,326	34.8%
Fiber-to-the-Premises	630	16.6%
All Technologies	3,808	100%

BEAD Outlay Reduced Due to Provider Cost Sharing:



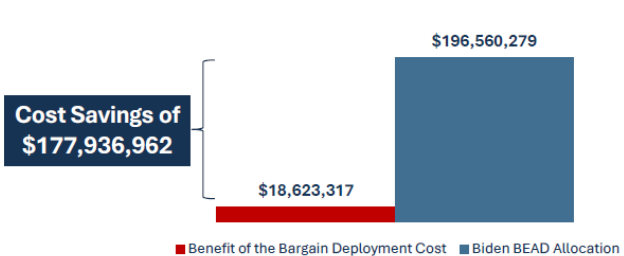
Avg. BEAD Cost per Location: \$4,946
Avg. Match per Location: \$2,471

Note that monetary values shown on this page are rounded to the nearest dollar.

BEAD Final Proposal New Hampshire Overview

Total Allocation: \$196,560,279
Total Deployment Cost: \$18,623,317
of Eligible Broadband Serviceable Locations: 5,135

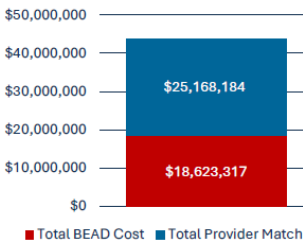
Benefit of the Bargain Saves Taxpayer Money:



Technology Mix Reflects the Competitive Marketplace:

Technology Type	# of Locations	% of Locations
Fiber-to-the-Premises	3,476	67.7%
Low Earth Orbit Satellite	1,104	21.5%
Hybrid Fiber-Coaxial	555	10.8%
All Technologies	5,135	100%

BEAD Outlay Reduced Due to Provider Cost Sharing:



Avg. BEAD Cost per Location: \$3,627
Avg. Match per Location: \$4,901

Note that monetary values shown on this page are rounded to the nearest dollar.



*Data based upon Final Proposal submissions and subject to change pending NTIA curing and NIST final review and award. Updated 11/07/2025.



*Data based upon Final Proposal submissions and subject to change pending NTIA curing and NIST final review and award. Updated 12/18/2025.



*Data based upon Final Proposal submissions and subject to change pending NTIA curing and NIST final review and award. Updated 11/13/2025.



SUB-AWARDEE FUNDING PROCESS

- Applicants must submit:

- Technical design (fiber, fixed wireless, etc.)
- Cost model & budget
- Matching funds documentation
- Cybersecurity & supply-chain compliance plans
- Workforce plan
- Environmental & permitting readiness
- Affordability commitments

Meeting Topic

- NTIA Special Award Condition (SAC) Review

- Before funds flow, NTIA conducts a SAC meeting with the state to verify:
- Compliance with BEAD rules
- Subgrantee selection integrity
- Cybersecurity & supply-chain requirements
- Budget and match validation
- Only after SAC approval can the state issue final awards.

Meeting Topic



**National Telecommunications and
Information Administration**

BEAD CYBER PLAN REQUIREMENTS

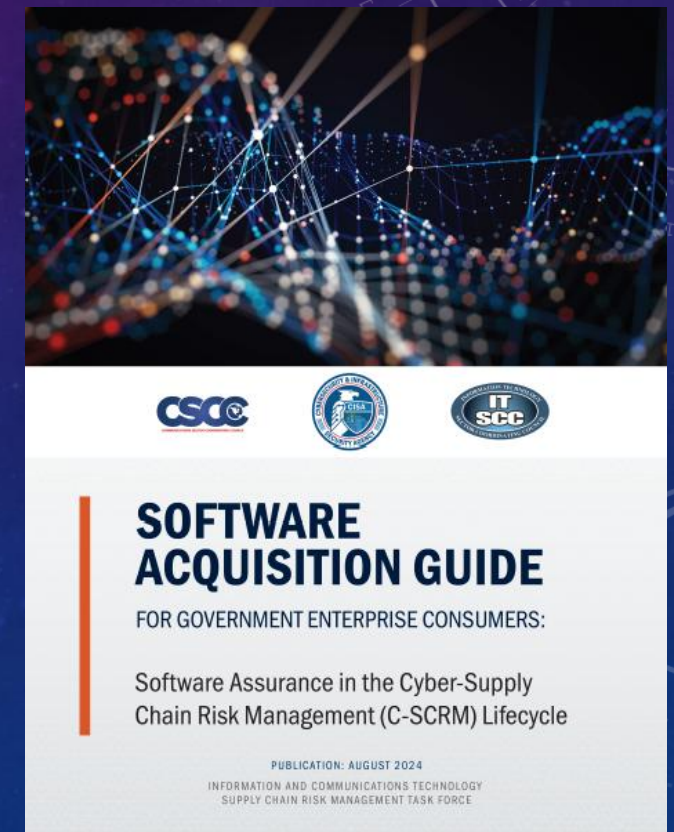
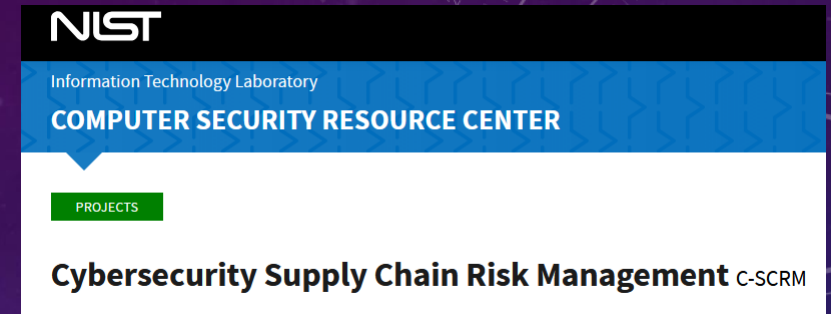
- **The Cybersecurity Risk Management Plan (CRMP)**
- A subgrantee must have a plan that is:
- **A. Operational or ready to be operationalized**
 - This is the *single most important gating requirement*.
- **B. Aligned with the NIST Cybersecurity Framework (CSF)**
 - Must reflect the **latest version** of the NIST CSF
 - This does **not** require full maturity or certification — only alignment.
- **C. Covers the CSF Core Functions**
 - At minimum, the plan must address:
 - **Governance**
 - **Identify** (assets, risks, roles, governance)
 - **Protect** (access control, training, data security)
 - **Detect** (anomalies, continuous monitoring)
 - **Respond** (incident response processes)
 - **Recover** (restoration, lessons learned)



[CISA CPG V2 alignment available now](#)

BEAD SCRM PLAN REQUIREMENTS

- **The Cyber Supply Chain Risk Management (C-SCRM) Plan**
- **Minimum Requirements**
- A subgrantee must have a plan that:
 - **A. Aligns with NIST 800-161 and NISTIR 8276**
 - NTIA explicitly points implementers to these documents as the baseline for SCRM.
 - **B. Addresses vendor and component (product) risk**
 - At minimum:
 - Identification of suppliers
 - Vetting of vendors
 - Controls for product procurement
 - Handling of foreign-manufactured equipment risks
 - Processes for monitoring supplier and product vulnerabilities
 - **C. Is operational or ready to operationalize**
 - Same gating rule as the cybersecurity plan.



GUIDANCE FOR BEAD IMPLEMENTERS

- BEAD Cybersecurity Requirements:
- <https://broadbandusa.ntia.doc.gov/sites/default/files/2022-05/BEAD%20NOFO.pdf> (p. 70)
- 1. Cybersecurity and supply chain risk management plans (the ICT SCRM Task Force created this guide from small businesses: https://www.cisa.gov/sites/default/files/2023-10/Empowering%20SMBs%20-%20A%20Resource%20Guide%20For%20Developing%20A%20Resilient%20SCRM%20Plan_508_0.pdf)
- a. NIST Framework for Improving Critical Infrastructure Cybersecurity^[1] (this [Quick Start Guide for small businesses](#) outlines the basics really well)
- b. Security and privacy controls being implemented from [Executive Order 14028](#)
- c. [NISTIR 8276, Key Practices in Cyber Supply Chain Risk Management: Observations from Industry](#)
- d. Supply chain risk management controls being implemented from [NIST 800-161, Cybersecurity Supply Chain Risk Management Practices for Systems and Organizations](#)
- 2. The plans must be submitted to the Eligible Entity prior to the allocation of funds.
- 3. BEAD providers must reevaluate and update the plans on a periodic basis and as events warrant;
- 4. If the BEAD provider makes any substantive changes to the plan(s), a new version will be submitted to the Eligible Entity within 30 days.
- 5. If network facilities are owned or operated by a third party (e.g., purchases wholesale carriage on such facilities), the BEAD provider must obtain the above attestations from its network provider.

NEED HELP

- Reach out to your State Broadband Administrators (Look on the BEAD Website for State Contact Info)
- <https://broadbandusa.ntia.gov/resources/states>
- BEAD Website <https://broadbandusa.ntia.gov/funding-programs/broadband-equity-access-and-deployment-bead-program>

HOW CAN CISA RESOURCES HELP



- CISA created several useful guides to help parties implement effective cyber and C-SCRM best practices for BEAD
- Cybersecurity and supply chain risk management plans (the ICT SCRM Task Force created this guide from small businesses: https://www.cisa.gov/sites/default/files/2023-10/Empowering%20SMBs%20-%20A%20Resource%20Guide%20For%20Developing%20A%20Resilient%20SCRM%20Plan_508_0.pdf)
- CISA Secure by Design Software Acquisition Guide (SAG) (covered in next section)
 - CISA SAG Website <https://cisa.gov/sag>
 - SAG Document: https://www.cisa.gov/sites/default/files/2024-07/PDM24050%20Software%20Acquisition%20Guide%20for%20Government%20Enterprise%20ConsumersV2_508c.pdf
 - SAG Spreadsheet: https://www.cisa.gov/sites/default/files/2024-08/PDM24064%20Software%20Acquisition%20Guide%20for%20Government%20Enterprise%20Consumers%20Final-%2020240710_v19.xlsx
 - SAG FACT Sheet: <https://www.cisa.gov/sites/default/files/2024-10/ICT%20SCRM%20Task%20Force%20Software%20Acquisition%20Guide%20Fact%20Sheet%20%28508%29.pdf>
 - SAG Webtool Self Assessment: <https://www.cisa.gov/software-acquisition-guide/tool>
 - SAG Overview Webinars: <https://www.youtube.com/watch?v=RzQMXE4Df0>

WHAT IS THE CISA SOFTWARE ACQUISITION GUIDE

- Aligns with CISA Secure by Design principles but focuses on the “Secure by Demand” elements.
- Identifies Secure by Design best practices across the entire software life cycle, based on NIST standards
- Enables acquisition professionals to engage in more relevant discussions with their enterprise risk owners (such as Chief Information Officers and Chief Information Security Officers) and candidate suppliers
- A virtual “Secure by Design” checklist for consumers and producers

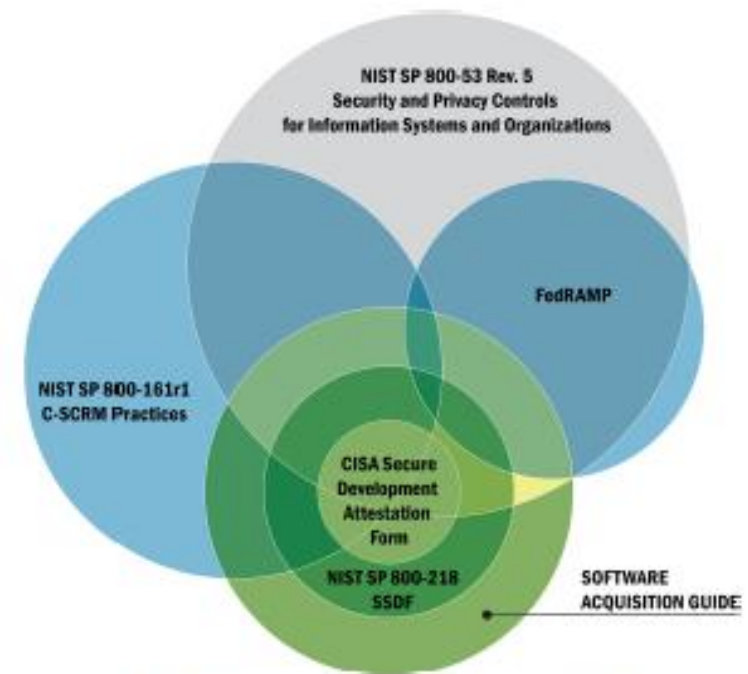


Figure 1. Notional overlap of major cybersecurity control efforts

[View the CISA NASPO Webinar for a comprehensive overview](#)

Software Acquisition Guide

This guide is organized into five primary sections with each section having its own set of controls and clarifying tasks, including:

- 19 CONTROL questions for Supplier Governance and Attestations
- 8 CONTROL questions for Software Supply Chain
- 30 CONTROL questions for Secure Software Development
- 12 CONTROL questions for Secure Software Deployment
- 8 CONTROL questions for Vulnerability Management

The questions are organized into a series of CONTROL questions with most CONTROL questions having a series of informative TASK questions. For each CONTROL question, it is expected that the software supplier will provide a simple response of “Yes,” “No,” “N/A,” or “Partial.”

WHO CREATED THE SAG MATERIALS

- CISA ICT_SCRM Task Force (CIPAC partnership)
- CISA National Risk Management Center
- Core group of ~30 people working on the SAG document
- Mostly from Government Agencies (NASA, GSA, CISA, NSA) and Private Sector <https://www.cisa.gov/ict-scrm-task-force-members>
- Led and facilitated by CISA personnel over two years
- Visionary leadership provided by an expert with decades of experience in software assurance, Joe Jarzombek, <https://copilot.microsoft.com/shares/QLhY366t3XSbYmfegpXEa>



ICT SCRM Task Force Members

The ICT SCRM Task Force is composed of a diverse range of representatives from large and small private sector organizations within the [Information Technology \(IT\)](#) and [Communications sectors](#), ICT associations, and federal agencies. Members include subject matter experts, ICT sector owners and operators, and other key stakeholders who provide recommendations and guidance to help shape trusted supply chain practices.

Companies and organizations participating in the Task Force include:

SECURE BY DESIGN, DEFAULT, DEMAND

- *“Secure by design’ means that technology products are built in a way that reasonably protects against malicious cyber actors successfully gaining access to devices, data, and connected infrastructure.”* (Source: Principles and Approaches, page 8) This sets an expectation for suppliers and includes how suppliers should take ownership of security outcomes for their customers, demonstrate radical transparency and accountability, and lead from the top by making secure by design a top business priority. This applies to the entire software development lifecycle (SDLC)/product lifecycle.
- *“Secure by default’ means products are resilient against prevalent exploitation techniques out of the box without added charge...without end-users having to take additional steps to secure them...make customers acutely aware when they deviate from safe defaults...”* (Source: Principles and Approaches, page 9) This means products come with a baseline level of security by default that do not need significant effort by customers to ‘harden’ the products and software against exploitation. The CISA guidance also states that, “The complexity of security configuration should not be a customer problem...,” (Source: Principles and Approaches, page 9) and that customers are not charged extra for implementing added security configurations.
- *‘Secure by demand’ means customer expectations for secure software and products are articulated in acquisition and procurement activities and contracts.* Within government agencies, employees performing requirements and contracting functions have significant roles in communicating enterprise expectations for secure software. Agencies can provide more complete guidance to support their efforts to focus on obtaining secure software. Since it remains challenging for suppliers to make the required security investments and efforts without associated demand, the updated CISA guidance also emphasizes the need for consumer demand: *“...just as we seek to create a pervasive secure by design philosophy within software manufacturers, we need to create a ‘secure by demand’ culture with their customers.”* (Source: Principles and Approaches, page 7) To focus on the need to correct market failures of cybersecurity, increased demand for secure products and software from enterprise customers and consumers is needed in acquisition and procurement because it is directly tied to customer demand and spending, which impacts supplier revenue and profits.

HOW DO PRODUCT MANUFACTURERS USE THE SAG

- Self Assessment of adherence to Secure by Design best practices in the SAG
- Perform self assessment before submitting a product to the US Government for procurement
- Answer the 19 Governance Questions, at a minimum
- Address any “red flags” where lack of adherence to Secure by Design could block a product from being purchased
- Create SAG Spreadsheet once and send to everyone that asks to show your Secure by Design practices
- Track Secure by Design practices per product

HOW DO CONSUMERS USE THE CISA SAG

- Assess a product and supplier for adherence to Secure by Design best practices
- Five step process
 1. Download the [CISA SAG Spreadsheet](#)
 2. Send the spreadsheet to your suppliers with a cover letter asking them to answer the 19 governance questions, at a minimum and return the completed spreadsheet
 3. Place the completed SAG Spreadsheets received in a permanent location for later review
 4. Evaluate each returned spreadsheet and decide if the provided responses are acceptable within your risk threshold and tolerance – [open source, free to use tools are available for this step](#)
 5. Keep a history of each analysis in a “log” with PASS/FAIL results – see open source tool above

POPULARITY



Reported metrics

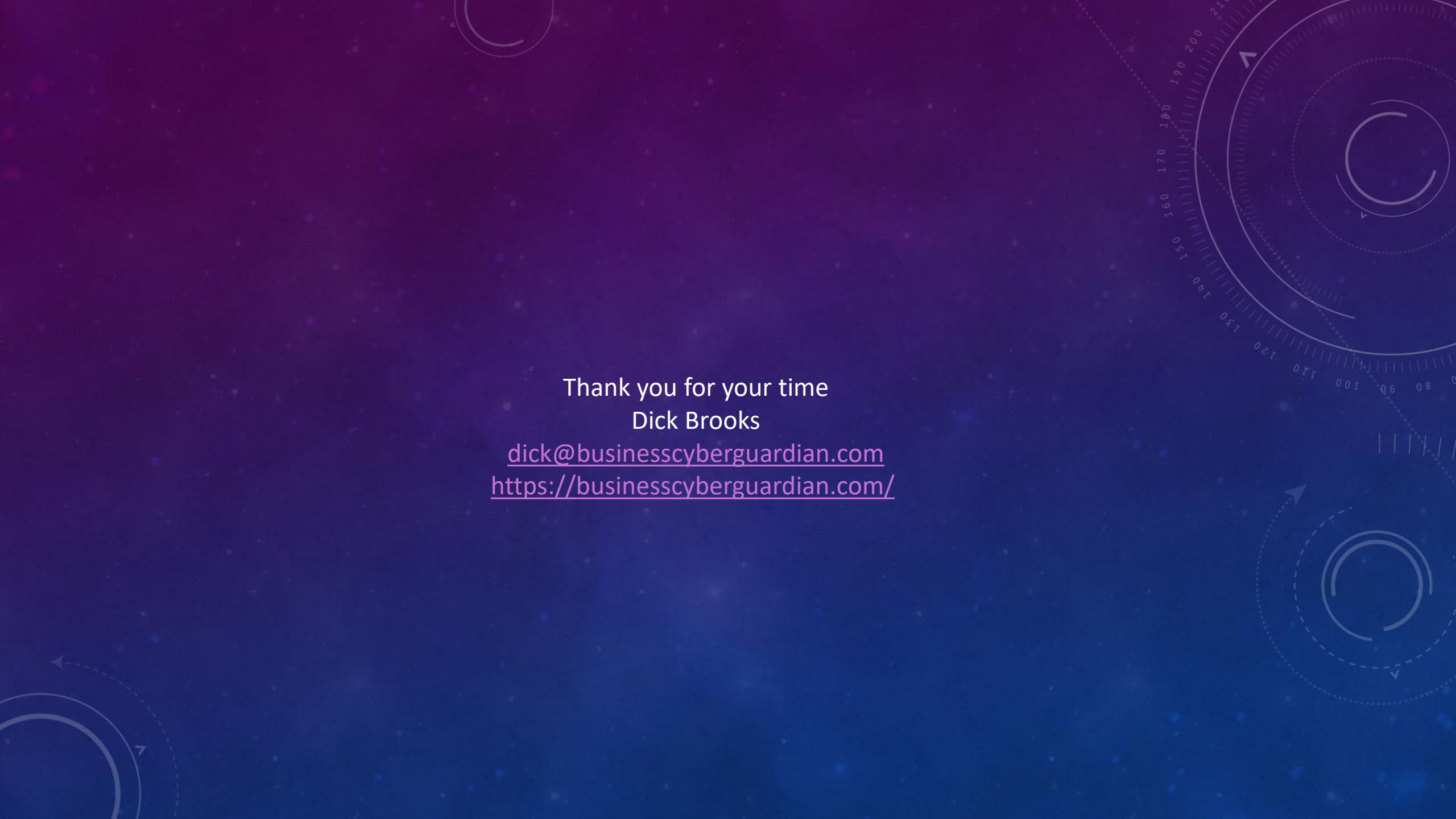
From the CISA SSCA Forum slide deck:

Resource	Reported Clicks (as of Jan 23, year not specified)
Software Acquisition Guide (PDF)	10,430
Software Acquisition Guide Spreadsheet	~5,000

Sources: [NIST Computer Security Resource Center](#)

RESOURCES

- **BEAD Resources**
 - <https://broadbandusa.ntia.gov/sites/default/files/2022-05/BEAD%20NOFO.pdf>
 - **BEAD Website** <https://broadbandusa.ntia.gov/funding-programs/broadband-equity-access-and-deployment-bead-program>
 - **BEAD Restructuring Policy Notice** https://broadbandusa.ntia.gov/funding-programs/policies-waivers/BEAD_Restructuring_Policy_Notice
 - <https://www.ntia.gov/press-release/2026/statement-assistant-secretary-roth-ntia-s-approach-use-21-billion-bead-savings>
 - **State BEAD Administrators:** <https://broadbandusa.ntia.gov/resources/states>
 - **NIST CSF V2.0 Guidance:** <https://www.nist.gov/cyberframework>
- **NTCA Resources**
 - <https://www.ntca.org/advocacy/issues/cybersecurity>
- **NASA SCRM Process Documentation**
 - <https://www.nasa.gov/secure-software-development-self-attestation-resources-and-knowledge/>
- **CISA SAG Videos**
 - **NASPO Webinar:** <https://www.youtube.com/watch?v=RzQMXE4Df0>
 - **Tom Fanning Keynote:** <https://www.youtube.com/watch?v=0iYITlQfi3A>
- **CISA Free Services and Open Source Tools**
 - <https://www.cisa.gov/resources-tools/resources/no-cost-cybersecurity-services-and-tools>
 - <https://github.com/rjb4standards/CISASAGReader/blob/main/README.md>
 - <https://www.cisa.gov/cybersecurity-performance-goals-2-0-cpg-2-0>
- **CISA SAG Website**
 - <https://cisa.gov/sag>
 - **SAG Document:** https://www.cisa.gov/sites/default/files/2024-07/PDM24050%20Software%20Acquisition%20Guide%20for%20Government%20Enterprise%20ConsumersV2_508c.pdf
 - **SAG Spreadsheet:** https://www.cisa.gov/sites/default/files/2024-08/PDM24064%20Software%20Acquisition%20Guide%20for%20Government%20Enterprise%20Consumers%20Final-%2020240710_v19.xlsx
 - **SAG FACT Sheet:** <https://www.cisa.gov/sites/default/files/2024-10/ICT%20SCRM%20Task%20Force%20Software%20Acquisition%20Guide%20Fact%20Sheet%20%28508%29.pdf>
- **LinkedIn SBOM SIG; Global Community of SBOM Implementers**
 - <https://www.linkedin.com/groups/13274064/>



Thank you for your time

Dick Brooks

dick@businesscyberguardian.com

<https://businesscyberguardian.com/>